TECHNICAL REPORT

1. Attack Summary

Three distinct security incidents were analyzed:

1. A malicious Microsoft Word document exploiting Follina (CVE-2022-30190) to achieve remote code execution via msdt.exe.
2. A phishing email campaign impersonating Amazon, using obfuscation (Base64) and SafeLinks redirection to deliver a malicious URL.
3. A high-volume RDP brute force attack targeting the default administrator account using repeated authentication attempts from a single external IP.

All three incidents demonstrate different stages of the cyber kill chain: initial access (phishing), exploitation (Follina) and credential attack (brute force).

2. Timeline

Follina Exploit

- Malicious DOCX delivered
- User opens document
- WINWORD.exe spawns msdt.exe
- Remote HTML payload retrieved
- Command execution triggered

Phishing Attack

- Email sent to victim
- User receives “Account locked” message
- SafeLinks URL redirects to attacker-controlled domain
- Base64 content decoded reveals hidden resources

RDP Brute Force

- Attacker initiates repeated RDP login attempts
- 3103 failed logon events generated
- Continuous attempts against administrator
- All attempts fail (no successful login observed)

3. Indicators of Compromise (IOCs)

File-Based (Follina)

Type	Value
MD5	52945af1def85b171870b31fa4782e52

SHA1	06727ffda60359236a8029e0b3e8a0fd11c23313
SHA256	4a24048f81afbe9fb62e7a6a49adbd1faf41f266b5f9feecdceb567aec096784
File Type	DOCX (Office Open XML)

Email-Based (Phishing)

Type	Value
Sender	amazon@zyevantoby.cn
Recipient	saintington73@outlook.com
Subject	Your Account has been locked
Encoding	Base64

Network-Based

Type	Value
Malicious URL	Safelinks redirect to attacker domain
Source IP	113.161.192.227
Country	Vietnam
Port Range	49162–65534

Behavioral Indicators

Indicator	Description
WINWORD.exe → msdt.exe	Follina exploitation pattern
Event ID 4625	Failed logon attempts
Repeated login failures	Brute force attack behavior

4. Tools Used

Tool	Purpose
VirusTotal	File hash and detection analysis
ANY.RUN	Dynamic malware behavior analysis
CyberChef	Base64 decoding and data extraction
Text Editor (Sublime)	Email header and content inspection
Log analysis tools	Windows Event log investigation

5. Findings

5.1 Follina Exploit

- The document successfully leverages MSDT for execution without macros

- Strong vendor detection confirms malicious intent
- Uses legitimate system binaries (LOLBins) for execution

5.2 Phishing Attack

- Attacker impersonates Amazon to create urgency
- SafeLinks used to disguise malicious destination
- Base64 encoding hides embedded URLs and resources
- Likely objective: credential harvesting

5.3 RDP Brute Force Attack

- 3103 failed login attempts indicate automated attack
- Targeted default administrator account
- Single external IP repeatedly attempts access
- Use of high ephemeral ports suggests scripted attack
- No successful compromise observed

Overall Assessment

- Threat Level: High
- Attack Diversity: Multi-vector (Email, Exploit, Network)
- Primary Risks:
    - Remote code execution
    - Credential compromise
    - Unauthorized access attempts

These incidents collectively highlight weaknesses in:

- Email security controls
- Endpoint monitoring
- Authentication protections